AI-POWERED SUPPLY CHAIN DISRUPTION ASSISTANT

API Documentation

Complete REST API Reference

Base URL: http://127.0.0.1:8000 | API prefix: /api

Framework: FastAPI 0.115+ | Version: 2.0.0

Swagger UI: http://127.0.0.1:8000/docs

Overview

The FastAPI backend exposes RESTful JSON endpoints under the /api prefix. All responses are JSON. Request validation uses Pydantic; malformed or missing fields return HTTP 422 with a structured detail array. Protected routes require Authorization: Bearer <token> issued on registration or login.

All sample request and response objects are illustrative — they demonstrate the correct schema based on the implementation but have not been verified by live execution in this documentation environment.

1. Authentication and Session Management

POST /api/auth/register

Creates a new user account with scrypt password hashing and issues a 24-hour session token. No authentication required.

Request body:

Field	Type	Required	Constraints
name	string	Yes	1–120 characters.
email	string	Yes	Valid email; normalized to lowercase.
password	string	Yes	8–256 characters.

Success HTTP 201:

```
{ "access_token": "<64-char-token>", "token_type": "bearer", "user": { "id": 1, "name": "Analyst",
"email": "analyst@example.com", "created_at": "2026-10-10T12:00:00" } }
```

Errors: HTTP 409 duplicate email; HTTP 422 validation failure.

POST /api/auth/login

Verifies credentials against scrypt hash; creates a new session. Request body: email (string), password (string). Success HTTP 200: same structure as /register. Error: HTTP 401 on wrong credentials.

POST /api/auth/logout

Revokes the current bearer token. Requires: Authorization: Bearer <token>. Response HTTP 200: { "ok": true }.

GET /api/auth/me

Returns public profile for the authenticated user. Requires auth. Response: { "id": integer, "name": string, "email": string, "created_at": datetime }.

2. System Health

GET /api/health

No authentication required. Returns: status (ok/unavailable), database, chroma, bm25, langgraph (available/not available), models (map: name → boolean), llm (provider or 'LLM not configured'), artifact_errors (component → error or null).

3. Disruption Records

GET /api/disruptions

No authentication required. Query parameters: disruption_type (string, optional), industry (string, optional), supplier_region (string, optional), disruption_severity (integer 1–5, optional), limit (integer 1–500, default 50), offset (integer >=0, default 0). Response: { "total": integer, "limit": integer, "offset": integer, "results": [...disruption objects...] }.

GET /api/disruptions/{disruption_id}

Returns a single record by MySQL auto-increment id. Error: HTTP 404 if not found.

4. Disruption Analysis (Full Workflow)

POST /api/disruptions/analyze

Requires authentication. Runs the 10-node LangGraph workflow. Request body:

Field	Type	Req?	Constraints
description	string	Yes	12–1200 chars; prompt-injection patterns rejected.
disruption_type	string	Yes	Factory Incident | Cyber Attack | Port Congestion | Labor Strike | Geopolitical | Natural Disaster
industry	string	Yes	Pharmaceuticals | Electronics | Consumer Goods | Automotive | Aerospace
supplier_tier	string	Yes	'1' | '2' | '3' | '4'
supplier_region	string	Yes	Asia-Pacific | Europe | South America | North America | Africa/Middle East
supplier_size	string	Yes	Large | Medium | Small
has_backup_supplier	boolean	Yes	true or false
disruption_severity	integer	Yes	1–5 inclusive
production_impact_pct	float	No	0–100; contextual only
revenue_loss_usd	float	No	0–1e12
response_type	string	No	One of five response strategy values
response_time_days	integer	No	0–5000

Response HTTP 200 top-level fields: analysis_id, classification, priority, similar_cases, predictions, response_analysis, recovery_analysis, resilience_analysis, recommendation, evidence, what_if_simulation, validation, evaluation (RAGAS), provider_tracking, llm_status.

Errors: HTTP 401 unauthenticated; HTTP 503 LangGraph/retrieval unavailable; HTTP 500 server error.

5. Standalone Retrieval

POST /api/retrieval/similar

No authentication required. Request body:

- query (string, 3–1200 chars, required).
- filters (object, optional) — RetrievalFilters: disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity (int or {$eq/$gte/$lte: int}), response_type.
- top_k (integer 1–50, default 10, optional).

Response: { "results": [...], "count": integer, "filters": object, "weights": { "semantic": 0.7, "keyword": 0.3 } }.

6. Machine Learning Predictions

POST /api/predictions

No authentication required. Request body: disruption_type, industry, supplier_tier, supplier_region, supplier_size, has_backup_supplier, disruption_severity (same constraints as /analyze). Response: { "predictions": { "production_impact": {"prediction": float, "unit": "%"}, "financial_loss": {"prediction": float, "unit": "USD"}, "recovery_time": {"prediction": float, "unit": "days"}, "permanent_supplier_change": {"prediction": float, "unit": "probability"} }, "explanations": { ...feature importances... }, "features_used": [...], "reliability": object }. Error: HTTP 503 if models not loaded.

GET /api/models

Returns model availability flags and stored training metrics from evaluation/model_metrics.json. No authentication required.

7. What-If Simulation

POST /api/analysis/what-if

Requires authentication. Request body: analysis_id (integer, optional — loads cases from a previous analysis owned by the user) OR cases (array, optional). Response: { "strategies": [{ "strategy": string, "case_count": integer, "median_recovery_days": float, "median_production_impact_pct": float, "median_revenue_loss_usd": float, "effectiveness": string, "advantages": [...], "disadvantages": [...], "trade_offs": [...] }], "case_count": integer, "disclaimer": string }.

8. Recommendations

POST /api/recommendations

Requires authentication. Query parameter: analysis_id (integer > 0, required). Returns the stored recommendation for the specified analysis (user must own the analysis). Error: HTTP 404 if not found or not owned.

9. Supplier Risk Intelligence

GET /api/suppliers/risk

No authentication required. Query parameters: region (string, optional), tier (string, optional), industry (string, optional), min_severity (integer 1–5, optional), limit (integer 1–200, default 50). Returns aggregated vulnerability scores, risk by industry and region, backup coverage, and prioritized risk segments.

10. Analysis History

Endpoint	Description
GET /api/history	Paginated, filterable history. Auth required. Filters: date_preset, date_from, date_to, provider, model, disruption_type, severity, industry, supplier_region, search, page, page_size.
GET /api/history/{id}	Single history item with full request and response data. Auth required.
DELETE /api/history/{id}	Delete one history item and associated data. Returns { "deleted": 1 }.
DELETE /api/history	Clear all history for authenticated user. Returns { "deleted": integer }.
GET /api/history/filters	Distinct filter values for the user's history.
GET /api/history/stats	Aggregate stats: total_analyses, successful_requests, fallback_requests, average_response_time_ms, provider_counts.

11. Analytics

Endpoint	Returns
GET /api/analytics/overview	total_disruptions, avg_production_impact, total_revenue_loss, avg_response_time, avg_recovery_time, backup_coverage_pct, permanent_change_rate_pct.
GET /api/analytics/disruptions	Disruption counts by type, industry, supplier_region, disruption_severity.
GET /api/analytics/responses	Median production impact, revenue loss, response days, partial/full recovery days per response_type.
GET /api/analytics/recovery	Full recovery time distribution bucketed (0–14, 15–30, 31–60, 61–90, 91–180, 181+ days) and overall median.

12. Feedback

POST /api/feedback (HTTP 201)

Requires authentication. Submits actual observed outcomes for a completed analysis.

Field	Type	Required	Notes
analysis_id	integer > 0	Yes	Must be owned by the authenticated user.
useful	boolean	Yes	Whether the recommendation was useful.
actual_response	string	No	One of five response type values.
actual_recovery_days	float 0–5000	No	Observed full recovery duration.
actual_production_impact_pct	float 0–100	No	Observed production impact.
actual_revenue_loss_usd	float 0–1e12	No	Observed revenue loss in USD.
comment	string max 1000	No	Free-text comment.

Response: { "id": integer, "message": "Feedback saved" }.

GET /api/feedback

Requires authentication. Full feedback analysis: stats, recurring_issues, improvement_insights, prediction_error_metrics (MAE drift), retraining_evaluation, error_trends.

GET /api/feedback/stats

Requires authentication. Summary: stats, recurring_issues, improvement_insights, prediction_error_metrics, retraining_evaluation, error_trends.

13. Evaluation

GET /api/evaluation

No authentication required. Returns model_metrics (from evaluation/model_metrics.json), evaluations (most recent 50 RAGAS records from MySQL), latest_report (evaluation/reports/latest_evaluation.json), evaluation_method ('RAGAS'), status ('available').

GET /api/evaluation/test-cases

Deprecated. Returns status 'deprecated' and message indicating RAGAS is the only evaluation method.

14. HTTP Status Code Reference

Status	Condition
200 OK	Successful GET or action request.
201 Created	Successful resource creation (POST /register, POST /feedback).
401 Unauthorized	Missing, expired, or invalid bearer token.
404 Not Found	Resource does not exist or is not owned by the user.
409 Conflict	Duplicate email address on registration.
422 Unprocessable Entity	Pydantic validation failure; detail array lists field path and message.
500 Internal Server Error	Unexpected server error; check backend logs for details.
503 Service Unavailable	LangGraph workflow, retrieval indexes, ML models not available, or DB unreachable.